

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2023 Worked Solutions

May/June 2023 paper rebuilt with the worked solution after each question.

11

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 May/June 2023

May/June 2023 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. The continuous curve C has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	4.0	4.2	4.4	4.6	4.8	5.0
y	9.2	8.4556	3.8512	5.0342	7.8297	8.6

Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_4^5 f(x) dx$$

giving your answer to 3 decimal places.

(3)

(Total for Question 1 is 3 marks)

(Total for Question 1 is 3 marks)

Worked Solution - Question 1

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 1 "0

Final answer

1 "0

Question 2

Polynomials

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

(a) use the factor theorem to show that $a = -3$

(2)

(b) Hence show that the equation $f(x) = 0$ has only one real root.

(4)

(Total for Question 2 is 6 marks)

(Total for Question 2 is 6 marks)

Worked Solution - Question 2

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: -no real roots so $3x^2 =$ is the only one real root * *

Final answer

-no real roots so $3x^2 =$ is the only one real root * *

Question 3

Circles

3. A circle C has centre $(2, 5)$

Given that the point $P(8, -3)$ lies on C

- (a) (i) find the radius of C

- (ii) find an equation for C

(3)

- (b) Find the equation of the tangent to C at P giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(Total for Question 3 is 7 marks)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: $-3.4360 \times 10^x y - - =$

Final answer

$-3.4360 \times 10^x y - - =$

Question 4

Binomial Expansion

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A , B , C and D are constants.

- (a) Find the value of A

(1)

Given that

- $B = 18D$
- $p < 0$

- (b) find

- (i) the value of p
(ii) the value of C

(6)

(Total for Question 4 is 7 marks)

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

Topic group

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: $\times \times - =$

Final answer

$\times \times - =$

Question 5

Laws of Logarithms

5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6)$$

(5)

(Total for Question 5 is 5 marks)

(Total for Question 5 is 5 marks)

Worked Solution - Question 5

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 6

Geometric Sequences

6. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$ (2)

(b) predict the number of downloads in the 4th month. (4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the N th month.

(c) Calculate the value of N according to the model. (3)

(Total for Question 6 is 9 marks)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: $21300 = c e$

Final answer

$$21300 = c e$$

Question 7

Applications of Differentiation

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river, H metres, t days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that H has a stationary value at $t = \alpha$

- (a) use calculus to show that α satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0 \quad (5)$$

- (b) Hence find the value of α , giving your answer to 3 decimal places. (1)

- (c) Use further calculus to prove that H is a maximum at this value of α . (2)

(Total for Question 7 is 8 marks)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: - - * *

Final answer

- - * *

Question 8

Proof

8. (i) A student writes the following statement:

“When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10”

Prove by counter example that this statement is **not** true.

(2)

- (ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

(Total for Question 8 is 5 marks)

(Total for Question 8 is 5 marks)

Worked Solution - Question 8

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: -170 is a multiple of 10 so the statement is untrue * * (ii) Check first if they have multiplied the inequalit

Final answer

-170 is a multiple of 10 so the statement is untrue * * (ii) Check first if they have multiplied the inequalit

Question 9

Trigonometric Equations

9. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0 \quad (3)$$

(b) Hence solve, for $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place.

(4)

(Total for Question 9 is 7 marks)

(Total for Question 9 is 7 marks)

Worked Solution - Question 9

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: =

Final answer

=

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10 marks

Question 10

Integration

10. The curve C has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x - k)^2}{\sqrt{x}} dx = ak^2 + bk + \frac{2046}{5}$$

where a and b are integers to be found.

(5)

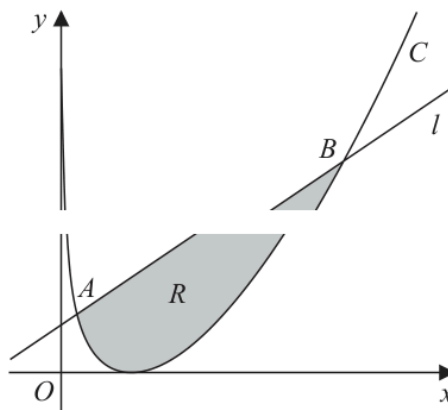


Figure 1

Figure 1 shows a sketch of the curve C and the line l .

Given that l intersects C at the point $A(1, 9)$ and at the point $B(16, q)$ where q is a constant,

(b) show that $k = 4$

(2)

The region R , shown shaded in Figure 1, is bounded by C and l

Using the answers to parts (a) and (b),

(c) find the area of region R

(3)

(Total for Question 10 is 10 marks)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 11

Sequences & Series

11. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where a and b are constants.

(a) Find, in terms of a and b ,

(i) u_2

(ii) u_3

(2)

Given

- $\sum_{n=1}^3 u_n = 153$

- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of u_2

(3)

(Total for Question 11 is 8 marks)

(Total for Question 11 is 8 marks)

Worked Solution - Question 11

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: $-\times- =$

Final answer

$-\times- =$